

ON SOME EXPECTATIONS WITH RESPECT TO WISHART MATRICES

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1. Introduction

This paper is concerned with some expectations of functions of the characteristic roots of a central Wishart (Σ, n) matrix $A(p \times p)$ and moments of A . It has been proved (Constantine (1963)):

$$1.1 \quad E C_K(A) = 2^{k(\frac{1}{2}n)} C_K(\Sigma)$$

where $C_K(A)$ is the zonal polynomial corresponding to partition $K = (k_1, k_2, \dots, k_p)$ of $K = \sum_{i=1}^p k_i$.

$$(\frac{1}{2}n)_K = \prod_{i=1}^p (\frac{1}{2}(n+1-i))_{k_i}$$

$$(a)_{k_i} = a(a+1) \dots (a+k_i-1)$$

Let $\text{tr}_k A$ denote the k -th elementary symmetric function (esf) of the roots of A , then it follows that

$$1.2 \quad \text{tr}_k A = (k+1)2^{-k} C_{1^k}(A)$$

Since (Constantine (1966))

$$1.3 \quad C_K(A) C_J(A) = \sum_D g_{K,J}^D C_D(A)$$

where $g_{K,J}^D$ is a constant tabulated by Khatri and Pillai (1968) for D a partition of $d = k+j = 1(1)7$, it follows that

$$1.4 \quad E(\text{tr}_k A \text{tr}_j A) = (k+1)(j+1) \sum_D g_{1k,1j}^D (\frac{1}{2}n)_D C_D(\Sigma)$$

and

$$1.5 \quad E(\text{tr}_k A \text{tr}_j A \text{tr}_h A) = (k+1)(j+1)(h+1) \sum_D g_{1k,1j}^D \sum_T g_{D,1}^T h(\frac{1}{2}n)_T C_T(\Sigma)$$

T being a partition of $k+j+h$.

2. On the expectations

The following expectations can now be derived using 1.2, 1.4 and 1.5.

$$2.1 \quad E \text{tr}_j A = n(n-1) \dots (n-j+1) \text{tr}_j \Sigma$$

$$2.2 \quad E \text{tr}_1 A \text{tr}_2 A = n(n-1)(n+2) \text{tr}_1 \Sigma \text{tr}_2 \Sigma - 6n(n-1) \text{tr}_3 \Sigma$$

$$2.3 \quad E \text{tr}_1 A \text{tr}_3 A = n(n-1)(n-2)(n+2) \text{tr}_1 \Sigma \text{tr}_3 \Sigma - 8n(n-1)(n-2) \text{tr}_4 \Sigma$$

$$2.4 \quad E \text{tr}_2^2 A = n(n+2)(n-1)(n+1) \text{tr}_2^2 \Sigma - 4n(n+2)(n-1) \text{tr}_1 \Sigma \text{tr}_3 \Sigma \\ - 4n(n-1)(2n-5) \text{tr}_4 \Sigma$$

$$2.5 \quad E \text{tr}_1^2 A \text{tr}_2 A = n(n+2)(n+4)(n-1) \text{tr}_1^2 \Sigma \text{tr}_2 \Sigma - 4n(n-1)(n+2) \text{tr}_2^2 \Sigma \\ - 12n(n-1)(n+2) \text{tr}_1 \Sigma \text{tr}_3 \Sigma + 48n(n-1) \text{tr}_4 \Sigma$$

$$2.6 \quad E \text{tr}_1^2 A = n(n+2) \text{tr}_1^2 \Sigma - 4n \text{tr}_2 \Sigma$$

$$2.7 \quad E \text{tr}_1^3 A = n(n+2)(n+4) \text{tr}_1^3 \Sigma - 12n(n+2) \text{tr}_1 \Sigma \text{tr}_2 \Sigma + 24n \text{tr}_3 \Sigma$$

$$2.8 \quad E \text{tr}_1^4 A = n(n+2)(n+4)(n+6) \text{tr}_1^4 \Sigma - 24n(n+2)(n+4) \text{tr}_2 \Sigma \text{tr}_1^2 \Sigma \\ + 48n(n+2) \text{tr}_2^2 \Sigma + 96n(n+2) \text{tr}_1 \Sigma \text{tr}_3 \Sigma - 192n \text{tr}_4 \Sigma.$$

The last three moments can also be found using the moment generating function of $\text{tr}_1 A$, i.e.

$$M(t) = |1-2t\Sigma|^{-n/2}.$$

Let $c_i (i = 1, \dots, p)$ be the characteristic roots of A, then it can be shown that

$$2.9 \quad \sum_{i \neq j} c_i c_j = 2 \operatorname{tr}_2 A$$

$$2.10 \quad \sum_{i \neq j \neq k} c_i c_j c_k = 6 \operatorname{tr}_3 A$$

$$2.11 \quad \sum_i c_i^2 = \operatorname{tr}_1^2 A - 2 \operatorname{tr}_2 A$$

$$2.12 \quad \sum_i c_i^3 = \operatorname{tr}_1^3 A - 3 \operatorname{tr}_1 A \operatorname{tr}_2 A + 3 \operatorname{tr}_3 A$$

$$2.13 \quad \sum_{i \neq j} c_i^2 c_j = \operatorname{tr}_1 A \operatorname{tr}_2 A - 3 \operatorname{tr}_3 A$$

$$2.14 \quad \sum_{i \neq j \neq k} c_i^2 c_j c_k = 2 \operatorname{tr}_1 A \operatorname{tr}_3 A - 8 \operatorname{tr}_4 A$$

$$2.15 \quad \sum_{i \neq j} c_i^2 c_j^2 = 2 \operatorname{tr}_2^2 A - 4 \operatorname{tr}_1 A \operatorname{tr}_3 A + 4 \operatorname{tr}_4 A$$

$$2.16 \quad \sum_i c_i^3 c_j = 4 \operatorname{tr}_4 A - \operatorname{tr}_1 A \operatorname{tr}_3 A - 2 \operatorname{tr}_2^2 A - 2 \operatorname{tr}_2^2 A + \operatorname{tr}_1^2 A \operatorname{tr}_2 A$$

$$2.17 \quad \sum_i c_i^4 = \operatorname{tr}_1^4 A + 2 \operatorname{tr}_2^2 A - 4 \operatorname{tr}_4 A + 4 \operatorname{tr}_1 A \operatorname{tr}_3 A - 4 \operatorname{tr}_1^2 A \operatorname{tr}_2 A.$$

Using 2.1 - 2.17 we have

$$2.18 \quad E \sum_i c_i = n \operatorname{tr}_1 \Sigma$$

$$2.19 \quad E \sum_{i \neq j} c_i c_j = 2n(n-1) \operatorname{tr}_2 \Sigma$$

$$2.20 \quad E \sum_{i \neq j \neq k} c_i c_j c_k = 6n(n-1)(n-2) \operatorname{tr}_3 \Sigma$$

$$2.21 \quad E \sum_{i \neq j \neq k \neq h} c_i c_j c_k c_h = 24n(n-1)(n-2)(n-3) \operatorname{tr}_4 \Sigma$$

$$2.22 \quad E \sum_i c_i^2 = n(n+2) \operatorname{tr}_1^2 \Sigma - 2n(n+1) \operatorname{tr}_2 \Sigma$$

$$2.23 \quad E \sum_i c_i^3 = n(n+2)(n+4) \operatorname{tr}_1^3 \Sigma - 3n(n+2)(n+3) \operatorname{tr}_1 \Sigma \operatorname{tr}_2 \Sigma \\ + 3n(n^2+3n+4) \operatorname{tr}_3 \Sigma$$

$$2.24 \quad E \sum_{i \neq j} c_i^2 c_j = n(n-1)(n+2) \operatorname{tr}_1 \Sigma \operatorname{tr}_2 \Sigma - 3n^2(n-1) \operatorname{tr}_3 \Sigma$$

$$2.25 \quad E \sum_{i \neq j \neq k} c_i^2 c_j c_k = 2n(n+2)(n-1)(n-2) \operatorname{tr}_1 \Sigma \operatorname{tr}_3 \Sigma \\ - 8n(n-1)^2(n-2) \operatorname{tr}_4 \Sigma$$

$$2.26 \quad E \sum_{i \neq j} c_i^2 c_j^2 = 2n(n+2)(n-1)(n+1) \operatorname{tr}_2^2 \Sigma - 4n^2(n-1)(n+2) \operatorname{tr}_1 \Sigma \operatorname{tr}_3 \Sigma \\ + 4n^2(n-1)^2 \operatorname{tr}_4 \Sigma$$

$$2.27 \quad E \sum_{i \neq j} c_i^3 c_j = n(n+2)(n+4)(n-1) \operatorname{tr}_1^2 \Sigma \operatorname{tr}_2 \Sigma - 2n(n+2)(n-1)(n+3) \operatorname{tr}_2^2 \Sigma \\ - n(n-1)(n+2)^2 \operatorname{tr}_1 \Sigma \operatorname{tr}_3 \Sigma + 4n(n-1)(n^2 + n + 4) \operatorname{tr}_4 \Sigma$$

$$2.28 \quad E \sum_i c_i^4 = n(n+2)(n+4)(n+6) \operatorname{tr}_1^4 \Sigma - 4n(n+2)(n+4)(n+5) \operatorname{tr}_1^2 \Sigma \operatorname{tr}_2 \Sigma \\ + 4n(n+2)(n^2 + 7n + 16) \operatorname{tr}_1 \Sigma \operatorname{tr}_3 \Sigma + 2n(n+2)(n+3)(n+5) \operatorname{tr}_2^2 \Sigma \\ - 4n(n^3 + 6n^2 + 21n + 20) \operatorname{tr}_4 \Sigma$$

3. On the moments of A

The moment generating function of A is given by

$$3.1 \quad M_A(\Theta) = |I - 2\Theta\Sigma|^{-n/2} \quad (\text{Anderson (1958)})$$

where Θ is a symmetric matrix.

If Θ is regarded as a non-symmetric matrix, the moment generating function of A can be written as

$$3.2 \quad M_A(\Theta) = |I - (\Theta + \Theta')\Sigma|^{-n/2}$$

This moment generating function is differentiated with respect to the elements of Θ by means of Bargmann's methods. (Bargmann (a)).

For reference a few formulae regarding the differentiation are stated here:

$$3.3 \quad \frac{\partial X}{\partial x_{ij}} = E_{ij} \text{ where } E_{ij} \text{ is a matrix of the same order as } X = [x_{ij}](p \times p), \text{ but with}$$

zeros in all positions except a one in the ij -position.

$$3.4 \quad \text{tr}(E_{ij} A) = a_{ji}$$

$$3.5 \quad \text{tr}(E_{ij} A E_{k\ell} B) = a_{jk} b_{\ell i}$$

$$3.6 \quad \text{tr}(E_{ij} A E_{k\ell} B E_{mn} C) = a_{jk} b_{\ell m} c_{ni}$$

$$3.7 \quad \text{tr}(E_{ij} A E_{k\ell} B E_{mn} C E_{pq} D) = a_{jk} b_{\ell m} c_{np} d_{qi}$$

$$3.8 \quad \frac{\partial \text{tr} AXB}{\partial X} = A'B'$$

$$3.9 \quad \frac{\partial |X|}{\partial X} = |X|(X')^{-1}$$

$$3.10 \quad \frac{\partial Y^{-1}}{\partial x} = -Y^{-1} \frac{\partial Y}{\partial x} Y^{-1}$$

3.11 If y is a scalar function of the matrix Z which is a matrix function of the matrix X , then:

$$\frac{\partial y}{\partial x_{ij}} = \text{tr} \left(\frac{\partial y}{\partial Z} \cdot \frac{\partial Z'}{\partial x_{ij}} \right)$$

Differentiation of $M_A(\Theta)$ to Θ_{ij} yields

$$3.12 \quad \frac{\partial M_A(\Theta)}{\partial \Theta_{ij}} = \left(-\frac{n}{2} \right) |I - (\Theta + \Theta') \Sigma|^{-\frac{n}{2} - 1} \text{tr} \left[\frac{\partial |I - (\Theta + \Theta') \Sigma|}{\partial (1 - (\Theta + \Theta') \Sigma)} \frac{\partial (1 - \Sigma(\Theta + \Theta'))}{\partial \Theta_{ij}} \right]$$

when using 3.10 and 3.11

$$= \frac{n}{2} |I - (\Theta + \Theta') \Sigma|^{-n/2} \text{tr} [(1 - \Sigma(\Theta + \Theta'))^{-1} \Sigma (E_{ij} + E_{ji})]$$

so that

$$\begin{aligned} E(a_{ij}) &= \frac{\partial M_A(\Theta)}{\partial \Theta_{ij}} \Big|_{\Theta=0} = \frac{n}{2} \text{tr} [(E_{ij} + E_{ji}) \Sigma] \\ &= n \sigma_{ij}, \text{ since } \Sigma \text{ is symmetric. Therefore} \end{aligned}$$

$$3.13 \quad E(A) = n \Sigma$$

Put $Y = (\Sigma^{-1} - (\Theta + \Theta'))^{-1}$ and differentiate 3.12 to $\Theta_{k\ell}$ which yields:

$$3.14 \quad \frac{\partial^2 M_A(\Theta)}{\partial \Theta_{k\ell} \partial \Theta_{ij}} = M_A(\Theta) \left\{ \frac{n}{2} \text{tr}(E_{ij} Y + E_{ji} Y) \frac{n}{2} \text{tr}(E_{k\ell} Y + E_{\ell k} Y) \right. \\ \left. + \frac{n}{2} \text{tr}(E_{ij} Y(E_{k\ell} + E_{\ell k}) Y + E_{ji} Y(E_{k\ell} + E_{\ell k}) Y) \right\}$$

when using 3.8, 3.9, 3.10 and 3.11.

Equate Θ to $O(p \times p)$ and use 3.5, then:

$$E(a_{ij} a_{k\ell}) = n^2(\sigma_{ij} \sigma_{k\ell}) + n(\sigma_{jk} \sigma_{i\ell} + \sigma_{j\ell} + \sigma_{j\ell} \sigma_{ik})$$

If $k = j$ and j is summated from 1 to p it follows

$$3.16 \quad E(A^2) = n^2 \Sigma^2 + n(\Sigma \text{tr} \Sigma + \Sigma^2) = n \Sigma \{ (n+1)\Sigma + I_p \text{tr} \Sigma \}$$

Differentiation of 3.14 to Θ_{mn} and equating Θ to $O(p \times p)$, k to j and m to ℓ gives:

$$3.17 \quad E(a_{ij} a_{j\ell} a_{\ell n}) = n^3 \sigma_{ij} \sigma_{j\ell} \sigma_{\ell n} + n^2(\sigma_{i\ell} \sigma_{\ell n} \sigma_{jj} + \sigma_{ij} \sigma_{j\ell} \sigma_{\ell n} \\ + \sigma_{in} \sigma_{j\ell} \sigma_{j\ell} + \sigma_{i\ell} \sigma_{\ell j} \sigma_{jn} + \sigma_{ij} \sigma_{jn} \sigma_{\ell\ell} + \sigma_{ij} \sigma_{j\ell} \sigma_{\ell n}) \\ + n(\sigma_{i\ell} \sigma_{\ell j} \sigma_{jn} + \sigma_{i\ell} \sigma_{\ell j} \sigma_{jn} + \sigma_{in} \sigma_{jj} \sigma_{\ell\ell} + \sigma_{i\ell} \sigma_{\ell n} \sigma_{jj} \\ + \sigma_{ij} \sigma_{j\ell} \sigma_{\ell n} + \sigma_{ij} \sigma_{jn} \sigma_{\ell\ell} + \sigma_{in} \sigma_{j\ell} \sigma_{j\ell} + \sigma_{i\ell} \sigma_{\ell j} \sigma_{jn})$$

When j and ℓ are summated from 1 to p it follows that:

$$3.18 \quad E(A^3) = n \Sigma \{ (n^2 + 3n + 4)\Sigma^2 + 2(n+1) \Sigma \text{tr} \Sigma + (n+1) I_p \text{tr} \Sigma^2 + I_p \text{tr}^2 \Sigma \}$$

Differentiate 3.14 to Θ_{sq} , equate Θ to $O(p \times p)$, k to j , m to ℓ , s to n and summate j , ℓ and n from 1 to p . The fourth moment of A then follows as:

$$3.19 \quad E(A^4) = n \Sigma \{ (n^3 + 6n^2 + 21n + 20)\Sigma^3 + (3n^2 + 10n + 12)\Sigma^2 \text{tr} \Sigma \\ + (2n^2 + 5n + 5)\Sigma \text{tr} \Sigma^2 + 3(n+1)\Sigma \text{tr}^2 \Sigma + (n^2 + 2n + 4)\text{tr} \Sigma^3 \\ + 3(n+1) I_p \text{tr} \Sigma \text{tr} \Sigma^2 + I_p \text{tr}^3 \Sigma \}$$

If the trace is taken of each of the results in 3.13, 3.16, 3.18 and 3.19 the results of equations 2.18, 2.22, 2.23 and 2.28 are obtained.

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